Problems sheet -Physics II (EBA1104)

Heat & Thermal Properties of Fluids

Textbook: Physics for Scientists and Engineers with Modern Physics

Serway & Jewett 10th Edition

Miscellaneous Exercises

Chapter 18: Temperature

- **Definitions:**

Two objects are in **thermal equilibrium** with each other if they do not exchange energy when in thermal contact.

Temperature is the property that determines whether an object is in thermal equilibrium with other objects. Two objects in thermal equilibrium with each other are at the same temperature. The SI unit of absolute temperature is the **kelvin,** which is defined to be 1/273.16 of the difference between absolute zero and the temperature of the triple point of water.

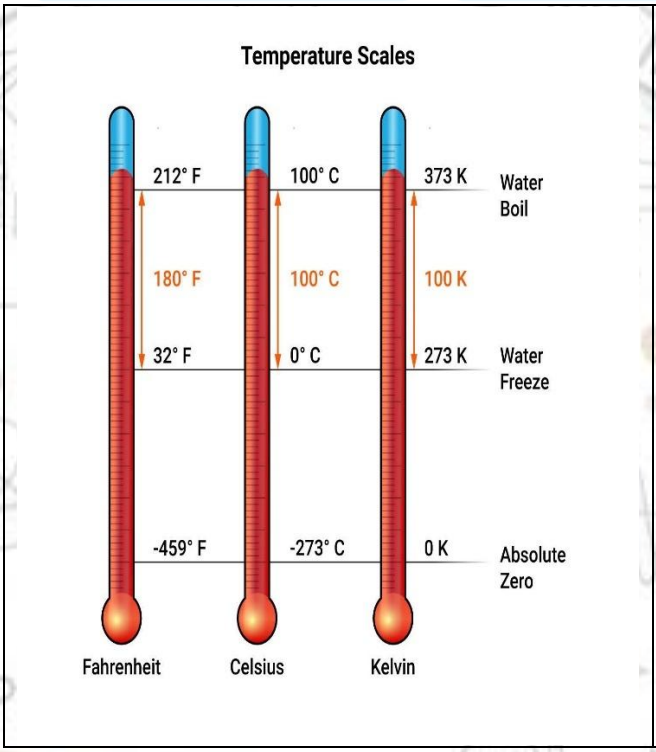
- **Concepts and Principles:**

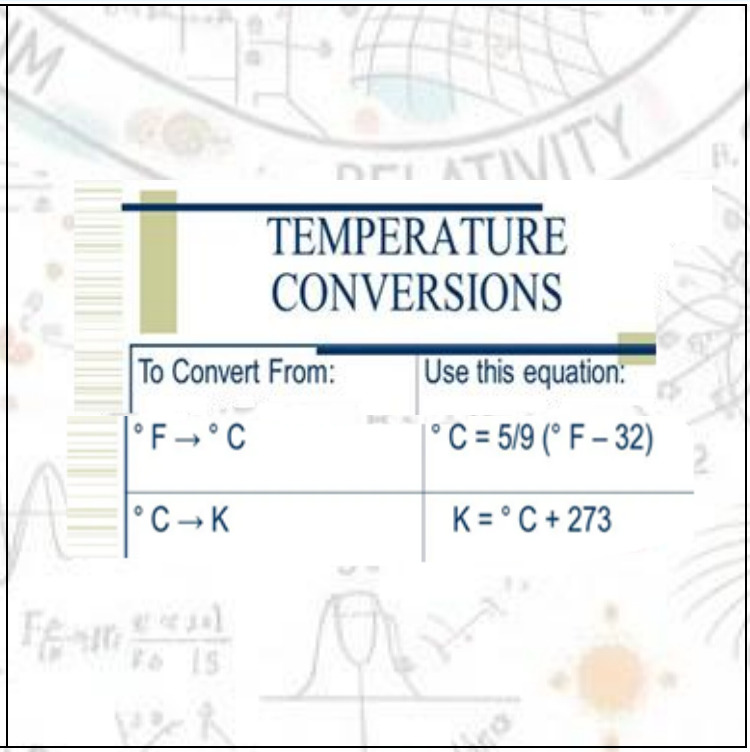
The **zeroth law of thermodynamics** states that if objects A and B are separately in thermal equilibrium with a third object C, then objects A and B are in thermal equilibrium with each other.

When the temperature of an object is changed by an amount ΔT, its length changes by an amount ΔL that is proportional to ΔT and to its initial length L_i:

$$\Delta L = \alpha L_i \Delta T \tag{18.5}$$

where the constant α is the **average coefficient of linear expansion.** The **average coefficient of volume expansion** β for a solid is approximately equal to 3α.





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Expansion of Materials

$\Delta L = \alpha L_0 \Delta T$ Linear Expansion

α = *Coefficient of Linear Expansion (solids)*

$\Delta V = \beta V_0 \Delta T$ Volume Expansion

β = *Coefficient of Volume Expansion*

$\beta = 3\alpha$ *(for solids)*

Chapter (18) Temperature:

a- **Celsius, Fahrenheit, and Kelvin temperature scales.**

1- A nurse measures the temperature of a patient to be 107°F. (a) What is this temperature on the Celsius scale? (b) Do you think the patient is seriously ill? Explain

2- The temperature difference between the inside and the outside of a home on a cold winter day is $57.0^{\circ}F$. Express this difference on (a) the Celsius scale and (b) the Kelvin scale.

3- Convert the following temperatures to their values on the Fahrenheit and Kelvin scales: (a) the sublimation point of dry ice, $-78.5^{\circ}C$; (b) human body temperature, 37.0 C.

4- The boiling point of liquid hydrogen is 20.3 K at atmospheric pressure. What is this temperature on (a) the Celsius scale and (b) the Fahrenheit scale?

5- A thermometer registers a change in temperature of 100°F. What change in temperature does this correspond to the Kelvin scale.

6-On a day when the temperature reaches 50F, what is the temperature in degrees Celsius and in kelvins?

7- Death Valley holds the record for the highest recorded temperature in the United States. On July 10, 1913, at a place called Furnace Creek Ranch, the temperature rose to 134F. The lowest U.S. temperature ever recorded occurred at Prospect Creek Camp in Alaska on January 23, 1971, when the temperature plummeted to 279.88F. (a) Convert these temperatures to the Celsius scale. (b) Convert the Celsius temperatures to Kelvin.

- **Choose the correct answer:**

1. Temperature is a property which determines
   a. how much heat a body contains
   b. whether a body will feel hot or cold to touch
   c. in which direction heat will flow between two systems
   d. how much total absolute energy a body has

2. The scales of temperature are based on two fixed points which are
   a. The temperatures of water at 0° 100°C
   b. The temperature of melting ice and boiling water at atmospheric pressure
   c. The temperatures of ice cold and boiling water
   d. The temperatures of frozen and boiling mercury

3. To understand the concept of temperature it isnecessary to understand
   a. the zeroth law of thermodynamics.
   b. the first law of thermodynamics.
   c. the second law of thermodynamics.
   d. all the above.
   e. only (b) and (c) above.

4. When two bodies are in thermal equilibrium with a third body, they are also in thermal equilibrium with each other. This statement is called
   a. Zeroth law of thermodynamics
   b. First law of thermodynamics
   c. Second law of thermodynamics
   d. Kelvin Planck's law

5. The temperature at which the volume of a gas becomes zero is called
   a. absolute scale of temperature
   b. absolute zero temperature
   c. absolute temperature
   d. none of these

6. The absolute zero temperature is taken as
   a. -273°C
   b. 273°C
   c. 237°F
   d. -237°F

7. The temperature of a body at 100°C is increased by ΔT as measured on the Celsius scale. How is this temperature change expressed on the Kelvin scale?
   a. ΔT
   b. ΔT + 273
   c. ΔT – 273
   d. ΔT + 373

8. In order for two objects to have the same temperature, theymust
   a. be in thermal equilibrium.
   b. be in thermal contact with each other.
   c. have the same relative "hotness" or "coldness"when touched.
   d. have all of the properties listedabove.
   e. have only properties (b) and (c) above.

9. A temperature difference of 5 K is equal to:
   a. a difference of 9 on the Celsius scale.
   b. a difference of 9 on the Fahrenheit scale.
   c. a difference of 2.8 on the Rankine scale.
   d. a difference of .5 on the Fahrenheit scale.
   e. a difference of 2.8 on the Celsius scale.

10. A thermometer registers a change in temperature of 100°F. What change in temperature does this correspond to on the Kelvin Scale?
   a. 453
   b. 328
   c. 180
   d. 55.6
   e. 24.5

11. Helium condenses into the liquid phase at approximately 4 K. What temperature, in degrees Fahrenheit, does thiscorrespond to?
   a. -182
   b. -269
   c. -118
   d. **-452**
   e. -484

12. Two thermometers are calibrated, one in degrees Celsius and the other in degrees Fahrenheit. At what temperature (in kelvins) do their readings measure the same temperature?

a. 218.15
b. 233.15
c. 273.15
d. 40.15
e. 0

13. A child has a temperature of 104°F. What is the temperature in degrees kelvin?

a. 40
b. 406
c. 401
d. 3 1 3
e . 3 4 9

14. At what temperature is the Celsius scale reading equal to twice the Fahrenheit scale reading?

a. **-12.3°F**
b. -24.6°F
c. -12.3°C
d. -6.1°C

15. A temperature difference of 9.0 Celsius degrees is equal to a Fahrenheit temperature difference of

a. 5.0 Fahrenheit degrees.
b. 9.0 Fahrenheit degrees.
c. 16 Fahrenheit degrees.
d. 37 Fahrenheit degrees.
e. 41 Fahrenheit degrees.

16. At what Fahrenheit temperature are the Kelvin and Fahrenheit temperatures numerically equal?

a. 574.6 °F
b. 220 °F
c. 150 °F
d. 100 °F

17. Death Valley in California receives many German tourists. When you convert a summer temperature of 130° F to the Celsius scale they use at home, you find that the Celsius temperature is about

a. 26° C.
b. **54° C.**
c. 72° C.
d. 176° C

18. A student has written the equation below to convert a temperature in degrees Fahrenheit into Kelvins. What is wrong with this equation?

$$T_K = \frac{9}{5} T_F + 32$$

a. The factor in front of T_F should be .
b. The numerical factor should multiply (T_F- 32).
c. An additional 273.15 Kelvins must be added to theright of the equation.
d. All the corrections above are required.
e. Only corrections (b) and (c) are required.

19. A temperature of 162° F is equivalent to what temperaturein kelvins?
a. 373 K
b. 288 K
c. 345 K
d. 201 K
e. 308 K

Chapter (18) Temperature:

b- Thermal expansion in solids and liquids

1- The concrete sections of a certain superhighway are designed to have a length of 25.0 m. The sections are poured and cured at $10.0°C$. What minimum spacing should the engineer leave between the sections to eliminate buckling if the concrete is to reach a temperature of $50.0°C$?

2- A copper telephone wire has essentially no sag between poles 35.0 m apart on a winter day when the temperature is $-20.0°C$. How much longer is the wire on a summer day when the temperature is 35.0 $°C$?

3- A pair of eyeglass frames is made of epoxy plastic. At room temperature ($20.0°C$), the frames have circular lens holes 2.20 cm in radius. To what temperature must the frames be heated if lenses 2.21 cm in radius are to be inserted in them? The average coefficient of linear expansion for epoxy is $1.30\times 10^{-4}C^{-1}$

4- The Trans-Alaska pipeline is 1300 km long, reaching from Prudhoe Bay to the port of Valdez. It experiences temperatures from $-73°C$ to $+35°C$. How much does the steel pipeline expand because of the difference in temperature? How can this expansion be compensated for? [α for steel = $11\times 10^{-6}C^{-1}$].

5- The active element of a certain LASER is made of a glass rod 30.0 cm long and 1.50 cm in diameter. Assume the average coefficient of linear expansion of the glass is equal to $9.00 \times 10^{-6}(℃)^{-1}$. If the temperature of the rod increases by 65.℃, what is the increase in (a) its length, (b) its diameter, and (c) its volume.

6- The average coefficient of volume expansion for carbon tetrachloride is 5.81×10^{-4} $(°C)^{-1}$. If a 50.0-gal steel container is filled completely with carbon tetrachloride when the temperature is 10.0°C, how much will spill over when the temperature rises to 30.0°C?

7- The 70-liter (L) steel gas tank of a car is filled to the top with gasoline at 20°C. The car sits in the Sun, and the tank reaches a temperature of 40°C (104°F). How much gasoline do you expect to overflow from the tank?

8- A building made with a steel structure is 650 m high on a winter day when the temperature is 0°F. What will the height (in cm) of the building be when it is 100°F? (The linear expansion coefficient of steel is $11\times 10^{-6}(°C)^{-1}$

- **Choose the correct answer:**

1- When the temperature of a rod of copper is increased, its length:

a. stays the same

b. increases

c. decreases

2- The amount by which unit length of a material increases when the temperature is raised one degree is called the coefficient of:

a. cubic expansion

b. superficial expansion

c. linear expansion

3- The symbol used for volumetric expansion is:

a. γ

b. β

c. L

d. A

4- A material of length L_1 at temperature T_1 K is subjected to a temperature rise of T K. The coefficient of linear expansion of the material is $\alpha\ K^{-1}$. The material expands by:

a. $L_2(1 + \alpha T)$

b. $L_1\alpha(T - T_1)$

c. $L_1[1 + \alpha(T - T_1)]$

d. $L_1\alpha T$

5- A bridge is made with segments of concrete 50 m long. If the linear expansion coefficient is 12×10^{-6} $(°C)^{-1}$, how much spacing (in cm) is needed to allow for expansion during an extreme temperature change of 150° F?

a. 10

b. 2.5

c. 7.5

d. 5.0

e. 9.5

6- A building made with a steel structure is 650 m high on a winter day when the

temperature is 0° F. How much taller (in cm) is the building when it is 100□ F? (The linear expansion coefficient of steel is 11×10^{-6} $(°C)^{-1}$.)

a. 71

b. 36

c. 40

d. 46

e. 65

7- A gallon container is filled with gasoline. How many gallons are lost if the temperature increases by 25°F? (The volume expansion of gasoline is 9.6×10^{-4} $(°C)^{-1}$.) (Neglectthe change in volume of the container.)

a. 2.4×10^{-2}

b. **1.3×10^{-2}**

c. 3.6×10^{-2}

d. 4.8×10^{-2}

e. 9.6×10^{-2}

8- A beaker is filled to the 500 ml mark with alcohol. What increase in volume (in ml) does the beaker contain when thetemperature changes from 5° C to 30° C? (Neglect theexpansion of the beaker, evaporation of alcohol and absorption of water vapor by alcohol.) $\beta_{alcohol} = 1.12 \times 10^{-4} °C^{-1}$

a. 0.47

b. 0.93

c. 1.4

d. 1.7

e. 2.5

9- A gold ring has an inner diameter of 2.168 cm at a temperature of 15.0°C. Determine its diameter at 100.0°C. ($\alpha_{GOLD} = 1.42 \times 10^{-5}/°C$)

a. 2.171cm

b. 2.171 m

c. 0.0217 cm

d. None of the above

10- Determine the change in length of a 20-m railroad track made of steel if the

temperature is changed from -15°C to +35°C. ($\alpha_{STEEL} = 1.1 \times 10^{-5}$/°C)

a. 0.011 cm
b. 4.4 mm
c. 1.1 cm
d. None of the above

11- A hole is drilled in a metal plate. When the metal is raised to a higher temperature, what happens to the diameter of the hole,

a. It decreases
b. It increases
c. It remains the same
d. The answer depends on the initial temperature ofthe metal.

12- Markings to indicate length are placed on a steel tape in a room that is at a temperature of 22°C. Measurements are then made with the same tape on a day when the temperature is 27°C. Assume the objects you are measuring have a smaller coefficient of linear expansion than steel. Are the measurements

a. too long,
b. too short,
c. accurate.

Chapter 19: The First law of Thermodynamics

- **Definitions:**

Internal energy is a system's energy associated with its temperature and its physical state (solid, liquid, gas). Internal energy includes kinetic energy of random translation, rotation, and vibration of molecules; vibrational potential energy within molecules; and potential energy between molecules.

Heat is the process of energy transfer across the boundary of a system resulting from a temperature difference between the system and its surroundings. The symbol Q represents the amount of energy transferred by this process.

A **calorie** is the amount of energy necessary to raise the temperature of 1 g of water from 14.5°C to 15.5°C.

The **heat capacity** C of any sample is the amount of energy needed to raise the temperature of the sample by 1°C.

The **specific heat** c of a substance is the heat capacity per unit mass:

$$c \equiv \frac{Q}{m\,\Delta T} \tag{19.3}$$

The **latent heat** of a substance is defined as the ratio of the energy input to a substance to the change in mass of the higher-phase material:

$$L \equiv \frac{Q}{\Delta m} \tag{19.7}$$

- **Concepts and Principles:**

The energy Q required to change the temperature of a mass m of a substance by an amount ΔT is

$$Q = mc\,\Delta T \tag{19.4}$$

where c is the specific heat of the substance.

The energy required to change the phase of a pure substance is

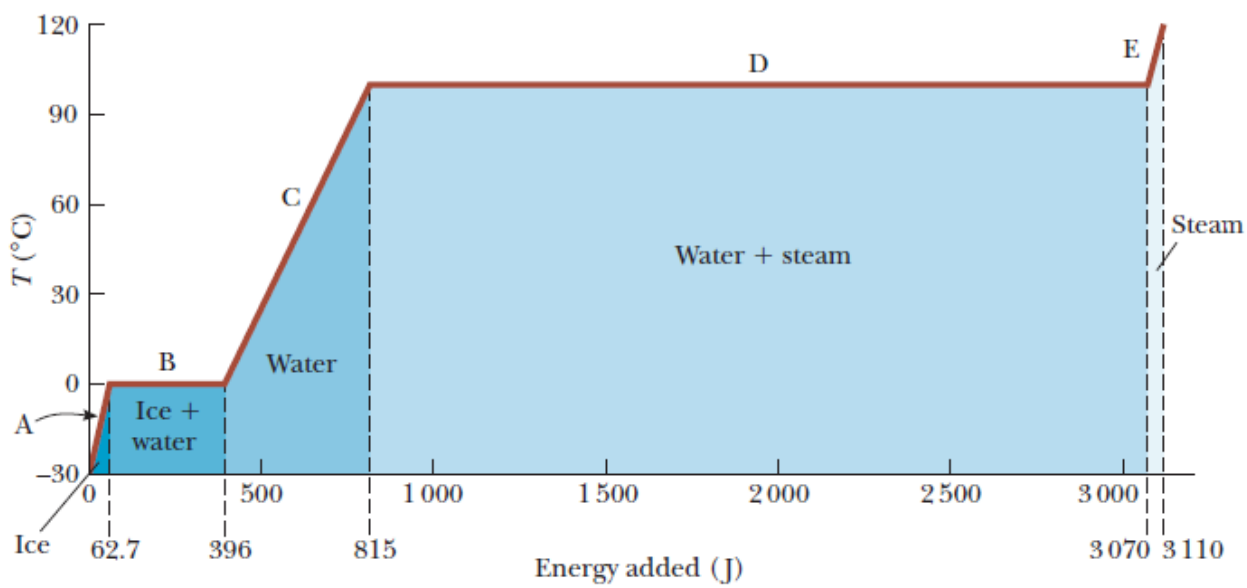
$$Q = L\,\Delta m \tag{19.8}$$

where L is the latent heat of the substance, which depends on the nature of the phase change and the substance, and Δm is the change in mass of the higher-phase material.

The **work** done on a gas as its volume changes from some initial value V_i to some final value V_f is

$$W = -\int_{V_i}^{V_f} P\,dV \tag{19.10}$$

where P is the pressure of the gas, which may vary during the process. To evaluate W, the process must be fully specified; that is, P and V must be known during each step. The work done depends on the path taken between the initial and final states.



Chapter (19) The First Law of Thermodynamics:

a- Specific Heat and Calorimetry

1- A combination of 0.250 kg of water at 20.0°C,0.40 kg of aluminum at 26.0°C, and 0.10 kg of copper at 100°C is mixed in an insulated container and allowed to come to thermal equilibrium. Ignore any energy transfer to or from the container. What is the final temperature of the mixture?

[Cwater = 4186, CAluminum = 900, Ccopper = 387 J/kg °C].

2- What mass of water at 25.0°C must be allowed to come to thermal equilibrium with a 1.85 kg cube of aluminum initially at 150°C to lower the temperature of the aluminum to 65.0°C?

Assume any water turned to steam subsequently condenses.

[Cwater = 4186, CAluminum = 900J/kg °C]

3- A 50.0 g sample of copper is at 25.0°C. If 1200 J of energy is added to it by heat, what is the final temperature of the copper?

[Ccopper = 387 J/kg °C].

4- An aluminum cup of mass 200 g contains 800 g of water in thermal equilibrium at 80.0°C. The combination of cup and water is cooled uniformly so that the temperature decreases by 1.50°C per minute. At what rate is energy being removed by heat? Express your answer in watts.

[Cwater = 4186, $C_{Almuinum}$= 900J/kg °C].

5- A 0.050 kg ingot of metal is heated to 200.0°C and then dropped into a calorimeter containing 0.40 kg of water initially at 20.0°C. The final equilibrium temperature of the mixed system is 22.4°C. Find the specific heat of the metal.

[Cwater = 4186J/kg °C].

6- The temperature of a silver bar rises by 10.08C when it absorbs 1.23 kJ of energy by heat. The mass of the bar is525 g. Determine the specific heat of silver from these data

7- An aluminum calorimeter with a mass of 100 g contains 250 g of water. The calorimeter and water are in thermal equilibrium at 10.0C. Two metallic blocks are placed into the

water. One is a 50.0-g piece of copper at 80.0C. The other has a mass of 70.0 g and is originally at a temperature of 100C. The entire system stabilizes at a final temperature of 20.08C. (a) Determine the specific heat of the unknown sample.

- **Choose the correct answer:**

1- The amount of heat required to raise the temperature of 1kg mass of that substance

through 1K is called its

a. specific energy
b. specific heat
c. specific stress
d. specific density

2- Heat absorbed or released by the same state of matter is represented

a. $Q = mc\Delta T$
b. $Q = mc/\Delta T$
c. $Q = m/c\Delta T$
d. $Q = \Delta T/mc$

3- Increase in temperature of the body is proportional to

a. amount of heat absorbed
b. amount of heat evolved
c. density of substance
d. average kinetic energy

4- The SI unit for thermal energy is

a. Joules
b. Ohms
c. Pascals
d. Watts

5- The standard symbol for heat capacity is

a. H
b. C
c. Q
d. E

6- The formula for calculating the heat capacity is

a. $H = \Delta T/Q$

b. $Q = C/\Delta T$
c. $Q = H/\Delta T$
d. $C = Q/\Delta T$

7- Formula for the specific heat capacity is
a. $c = C/\Delta T$
b. $c = Q/\Delta T$
c. $c = C/m$
d. $c = \Delta T/m$

8- SI unit of specific heat capacity is
a. Jkg^{-1}
b. K^{-1}
c. $Jkg^{-1} K^{-1}$
d. Pa

Chapter (19) The First Law of Thermodynamics:

b- Latent Heat

1- A 50.0 g copper calorimeter contains 250 g of water of 20.0°C. How much steam at 100°C must be condensed into water if the final temperature of the system reaches 50.0°C?

[Cwater = 4186, Ccopper = 387 J/kg °C, Lv = 2.26 × 106J/Kg].

2- A 75.0 g ice cube at 0°C is placed in 825 g of water at 25.0°C. What is the final temperature of the mixture

[Cwater = 4186 J/kg °C, Lf = 3.33 × 105J/Kg].

3- A 3.0 g lead bullet at 30.0°C is fired at a speed of 240 m/s into a large block of ice at 0°C, in which it becomes embedded. What quantity of ice melts? melts?

[Lf = 3.33 × 105J/Kg, Clead =128 J/kg °C]

4- A 1.00 kg block of copper at 20.0°C is dropped into a large vessel of liquid nitrogen at 77.3 K. How many kilograms of nitrogen will boil away by the time the copper reaches 77.3 K? (The specific heat of copper is 0.0920 cal/g. °C, and the latent heat of vaporization of nitrogen is 48.0 cal/g.)

5- In an insulated vessel, 250 g of ice at 0°C is added to 600 g of water at 18.0°C. (a) What is the final temperature of the system? (b) How much ice remains when the system reaches equilibrium?

[Cwater = 4186 J/kg °C, Lf = 3.33 × 105J/Kg].

6- Find the amount of heat required to convert 20 gm of water at 50 ℃ to steam at 120℃ If the specific heat capacity of water equal (4168 J/kg℃), the specific heat capacity of steam equal (2010 J/kg ℃), and the latent heat of vaporization of water is (2.26x10^6J/kg).

7- If 25 kg of ice at 0 °C is combined with 4 kg of steam at 100 °C, what will be the final equilibrium temperature (in °C) of the system?

8- How much heat must be removed to make ice at -10 °C from 2 kg of water at 20°C?

9- How much energy is required to change a 40.0-g ice cube from ice at -10.0°C to steam at 110°C?

10- What mass of steam initially at 1308C is needed to warm 200 g of water in a 100-g glass container from 20.0°C to 50.0°C?

- **Choose the correct answer:**

1- L_v is the symbol for

a. Latent heat of Fusion

b. Specific Latent Heat of Vaporization

c. Specific Latent Heat of Fusion
d. Latent Heat of Vaporization

2- L_f is the symbol for
a. Latent heat of Fusion
b. Specific Latent Heat of Vaporization
c. Specific Latent Heat of Fusion
d. Latent Heat of Vaporization

3- The heat capacity of sodium metal is 1500 JK^{-1}, if the mass of the sodium metal is 75 kg, the specific heat capacity would be
a. 10 J kg^{-1} $°C^{-1}$
b. 20 J kg^{-1} $°C^{-1}$
c. 40 J kg^{-1} $°C^{-1}$
d. 80 J kg^{-1} $°C^{-1}$

4- One Joule (J) is equal to
a. 1 N.m
b. 1 KN.m
c. 10 N.m/s
d. 10 KN.m/s

5- How many calories of heat are required to raise the temperature of 4 kg of water from 50°F to the boilingpoint?
a. 6.5×10^5
b. **3.6×10^5**
c. 15×10^5
d. 360
e. 4×10^4

6- A 5-gallon container of water (approximately 20 kg) having a temperature of 212°F is added to a 50-gallon tub (approximately 200 kg) of water having a temperature of 50°F. What is the final equilibrium temperature (in °C) of the mixture?
a. 54

b. 36
c. 18
d. 66
e. 14

7- A 5-kg piece of lead (specific heat 0.03 cal/g. °C) having a temperature of 80□C is added to 500 g of water having a temperature of 20°C. What is the final equilibrium temperature (in °C) of the system?
a. 79
b. 26
c. 54
d. 34

8- A 300-g glass thermometer initially at 25°C is put into 200 cm^3 of hot water at 95°C. Find the final temperature (in °C) of the thermometer, assuming no heat flows to the surroundings. (The specific heat of glass is 0.2 cal/g. °C.)
a. 52
b. 68
c. 89
d. 79
e. 36

9- How much heat (in kilocalories) is needed to convert 1.00 kg of ice at 0°C into steam at 100°C?
a. 23.9
b. 79.6
c. 564
d. 643
e. 720

10- If 25 kg of ice at 0°C is combined with 4 kg of steam at 100°C, what will be the final equilibrium temperature (in °C) of the system?
a. 40

b. 20

c. 60

d. 100

e. 8

11- How much heat (in kcal) must be removed to make ice at -10°C from 2 kg of water at 20°C? (The specific heat of iceis 0.50 cal/g. °C.)

a. 190

b. 200

c. 240

d. 210

12- How much heat, in joules, is required to convert 1.00 kg ofice at 0°Cinto steam at 100C? (L_{ice} = 333 J/g; L_{steam} = 2.26× 10^3 J/g.)

a. 3.35×10^5

b. 4.19×10^5

c. 2.36×10^6

d. 2.69×10^6

e. **3.01×10^6**

13- Water at room temperature, 20°C, is pumped into a reactor core where it is converted to steam at 200°C. How much heat (in J) is transferred to each kilogram of water in this process? (c_{steam} = 2 010 J/kg□□C; L_{steam} =2.26 × 10^3 J/g; 1 cal = 4.186 J.)

a. 3.35×10^5

b. 7.53×10^5

c. 2.67×10^6

d. **2.80×10^6**

e. 3.01×10^6

14- Steel blocks A and B, which have equal masses, are at T_A =300 °C and T_B = 400 °C. Block C, with $m_C = 2m_A$, is at T_C =350 °C. Blocks A and B are placed in contact, isolated, and allowed to come into equilibrium. Then they are placed in contact with block C. At that instant,

a. $T_A = T_B < T_C$.
b. **$T_A = T_B = T_C$.**
c. $T_A = T_B > T_C$.
d. $T_A + T_B = T_C$.
e. $T_A \times T_B = T_C$.

15- A poker is a stiff, nonflammable rod used to push burning logs around in a fireplace. For safety and comfort of use, should the poker be made from a material with

a. High specific heat and high thermal conductivity
b. Low specific heat and low thermal conductivity
c. Low specific heat and high thermal conductivity
d. High specific heat and low thermal conductivity

16- How much energy is required to raise the temperature of 5.00 kg of lead from 20.0° C to its melting point of 327°C? The specific heat if lead is 128 J/kg .°C.

a. 4.04×10^5 J
b. 1.07×10^5 J
c. 8.15×10^4 J
d. 2.13×10^4 J
e. 1.96×10^5 J

17- How long would it take a 1000 W heater to melt 1.00 kg of ice at – 20.0° C, assuming all the energy from the heateris absorbed by the ice?

a. 4.18 s
b. 41.8 s
c. 5.55 min
d. 6.25 min
e. 38.4 min